brane as it evolves through time? In short, might the universe as we know it be a brane?[4]

The possibility that we are living within a three-brane—the so-called *braneworld scenario*—is the latest twist in string/M-theory's story. As we will see, it provides a qualitatively new way of thinking about string/M-THEORY, with numerous and far-reaching ramifications. The essential physics is that branes are rather like cosmic Velcro; in a particular way we'll now discuss, they are very sticky.

Sticky Branes and Vibrating Strings

One of the motivations for introducing the term "M-theory" is that we now realize that "string theory" highlights but one of the theory's many ingredients. Theoretical studies revealed one-dimensional strings decades before more refined analyses discovered the higher-dimensional branes, so "string theory" is something of an historical artifact. But, even though M-theory exhibits a democracy in which extended objects of a variety of dimensions are represented, strings still play a central role in our current formulation. In one way this is immediately clear. When all the higher-dimensional *p*-branes are much heavier than strings, they can be ignored, as researchers had done unknowingly since the 1970s. But there is another, more general way in which strings are first among equals.

In 1995, shortly after Witten announced his breakthrough, Joe Polchinski of the University of California at Santa Barbara got to thinking. Years earlier, in a paper he had written with Robert Leigh and Jin Dai, Polchinski had discovered an interesting though fairly obscure feature of string theory. Polchinski's motivation and reasoning were somewhat technical and the details are not essential to our discussion, but his results are. He found that in certain situations the endpoints of open strings— remember, these are string segments with two loose ends —would not be able to move with complete freedom. Instead, just as a bead on a wire is free to move, but must follow the wire's contour, and just as a pinball is free to move, but must follow the contours of the pinball table's surface, the endpoints of an open string would be free to move but would be restricted to particular shapes or contours in space.